

Question 1

Let $C \subseteq \mathbb{R}^n$ be a convex set, with $x_1, \dots, x_k \in C$, and let $\theta_1, \dots, \theta_k \in \mathbb{R}_+$ satisfy $\theta_1 + \dots + \theta_k = 1$. Show that $\theta_1 x_1 + \dots + \theta_k x_k \in C$. (The definition of convexity is that this holds for $k = 2$; you must show it for arbitrary k .) *Hint.* Use induction on k .

Proof: for $m+1$: let $\tilde{\theta} = \sum_{i=1}^m \theta_i = 1 - \theta_{m+1}$

if $\tilde{\theta} = 0$, then $\theta_1 x_1 + \dots + \theta_{m+1} x_{m+1} = x_{m+1} \in C$

else,

there is $y = \sum_{i=1}^m \frac{\theta_i}{\tilde{\theta}} x_i \in C$

for $y, x_{m+1} \in C$,

$$\begin{aligned} \text{there is } & \theta_{m+1} \cdot x_{m+1} + (1 - \theta_{m+1})y \\ &= \sum_{i=1}^{m+1} \theta_i x_i \in C \end{aligned}$$

Question 2

Suppose A is an index set (e.g. $A = \{1, 2\}$, or $A = \{1, 2, \dots\}$) Assume that for each $\alpha \in A$, $C_\alpha \subset \mathbb{R}^n$ is convex, prove that $\bigcap_{\alpha \in A} C_\alpha$ is convex.

Proof: let $C = \bigcap_{\alpha \in A} C_\alpha$

$\forall x, y \in C, \Rightarrow$ for $\forall \alpha \in A, x \in C_\alpha, y \in C_\alpha$.

$\Rightarrow$ for $\forall \theta \in [0, 1], \theta x + (1 - \theta)y \in C_\alpha$

$\Rightarrow \theta x + (1 - \theta)y \in C$

$\Rightarrow \bigcap_{\alpha \in A} C_\alpha$ is convex

Question 3

Show that a set is convex if and only if its intersection with any line is convex.

① if set C is convex.

let $L \subset \mathbb{R}^n$, $L = \{x_0 + td \mid t \in \mathbb{R}\}$, $x_0 \in L$.

$$\forall x, y \in L, \quad x = x_0 + t_1 d, \quad y = x_0 + t_2 d$$

$$\Rightarrow \text{for } \forall \theta \in [0, 1], \quad \theta x + (1-\theta)y = \theta(x_0 + t_1 d) + (1-\theta)(x_0 + t_2 d) \\ = x_0 + [\theta t_1 + (1-\theta)t_2]d \in L$$

$$\Rightarrow L \text{ is convex.} \quad \Rightarrow C \cap L \text{ is convex.}$$

② $\forall x, y \in C$. Let L be the line through x, y .

$$\Rightarrow x, y \in C \cap L.$$

obviously $C \cap L$ is convex $\Rightarrow \forall \theta \in [0, 1], \theta x + (1-\theta)y \in C \cap L$

$$\Rightarrow \forall \theta \in [0, 1], \quad \theta x + (1-\theta)y \in C$$

since x, y are arbitrary in C .

$$\Rightarrow C \text{ is convex.}$$

Question 4

Which of the following sets are convex?

(a) A *slab*, i.e., a set of the form $\{x \in \mathbb{R}^n \mid \alpha \leq a^T x \leq \beta\}$.

$$\begin{aligned} & \{x \in \mathbb{R}^n \mid \alpha \leq a^T x \leq \beta\} \\ &= \{x \in \mathbb{R}^n \mid a^T x \geq \alpha\} \cap \{x \in \mathbb{R}^n \mid a^T x \leq \beta\} \\ &\Rightarrow \text{Intersection of two halfspaces} \Rightarrow \text{convex.} \end{aligned}$$

(b) A *rectangle*, i.e., a set of the form $\{x \in \mathbb{R}^n \mid \alpha_i \leq x_i \leq \beta_i, i = 1, \dots, n\}$. A rectangle is sometimes called a *hyperrectangle* when $n > 2$.

$$\begin{aligned} & \{x \in \mathbb{R}^n \mid \alpha_i \leq x_i \leq \beta_i\} \\ &= \{x \in \mathbb{R}^n \mid \alpha_1 \leq x_1 \leq \beta_1\} \cap \dots \cap \{x \in \mathbb{R}^n \mid \alpha_n \leq x_n \leq \beta_n\} \\ &\Rightarrow \text{Intersection of } 2n \text{ halfspaces} \Rightarrow \text{convex} \end{aligned}$$

(c) A *wedge*, i.e., a set of the form $\{x \in \mathbb{R}^n \mid a_1^T x \leq b_1, a_2^T x \leq b_2\}$.

Intersection of halfspaces $\Rightarrow$ convex

(d) The set of points closer to a given point than a given set, i.e.,

$$\{x \mid \|x - x_0\|_2 \leq \|x - y\|_2, \forall y \in S\}, \text{ where } S \subseteq \mathbb{R}^n.$$

$$\|x - x_0\|_2 \leq \|x - y\|_2$$

$$\Rightarrow (x - x_0)^2 \leq (x - y)^2$$

$$\Rightarrow x^2 - 2x_0x + x_0^2 \leq x^2 - 2xy + y^2$$

$$(2y - 2x_0)x \leq y^2 - x_0^2$$

$\Rightarrow$ halfspace is convex

(e) The set of points closer to one set than another, i.e.,

$$\{x \mid \text{dist}(x, S) \leq \text{dist}(x, T)\},$$

where $S, T \subseteq \mathbb{R}^n$, and

$$\text{dist}(x, S) = \inf \{\|x - z\|_2 \mid z \in S\}$$

Not always convex.

When $n=1$, $S = \{-1, 1\}$, $T = \{0\}$

$$\Rightarrow \{x \mid \text{dist}(x, S) \leq \text{dist}(x, T)\} =$$

$$\min \{\|x + 1\|_2, \|x - 1\|_2\} \leq \|x\|_2$$

$$\Rightarrow x \leq -\frac{1}{2} \text{ or } x \geq \frac{1}{2}$$

$\Rightarrow$ non-convex

(f) The set $\{x \mid x + S_2 \subseteq S_1\}$, where $S_1, S_2 \subseteq \mathbb{R}^n$ with S_1 convex.

$$\begin{aligned}\{x \mid x + S_2 \subseteq S_1\} &= \{x \mid x + y \in S_1, \forall y \in S_2\} \\ &= \bigcap_{y \in S_2} \{x \mid x + y \in S_1\} = \bigcap_{y \in S_2} \{x \mid x \in S_1 - y\}\end{aligned}$$

$\Rightarrow$ Translation and Intersection preserve convexity

(g) The set of points whose distance to a does not exceed a fixed fraction θ of the distance to b , i.e., the set $\{x \mid \|x - a\|_2 \leq \theta \|x - b\|_2\}$. You can assume $a \neq b$ and $0 \leq \theta \leq 1$.

$$\{x \mid \|x - a\|_2 \leq \theta \|x - b\|_2\} = \{x \mid (x - a)^2 \leq \theta^2 (x - b)^2\}$$

$$x^2 - 2ax + a^2 \leq \theta^2 x^2 - 2\theta^2 ax + \theta^2 b^2$$

$$\Rightarrow (1 - \theta^2)x^2 - 2a(1 - \theta^2)x + a^2 - \theta^2 b^2 \leq 0$$

if $\theta = 1$, it's a halfspace $\Rightarrow$ convex

else: $1 - \theta^2 > 0 \Rightarrow$ convex.

Question 5

What is the distance between two parallel hyperplanes $\{x \in \mathbb{R}^n \mid a^T x = b_1\}$ and $\{x \in \mathbb{R}^n \mid a^T x = b_2\}$?

$$\text{dist} = \frac{|b_1 - b_2|}{\|a\|_2}.$$

Question 6

Which of the following sets S are polyhedra? If possible, express S in the form $S = \{x \mid Ax \preceq b, Fx = g\}$.

(a) $S = \{y_1 a_1 + y_2 a_2 \mid -1 \leq y_1 \leq 1, -1 \leq y_2 \leq 1\}$, where $a_1, a_2 \in \mathbb{R}^n$.

Yes. $S = \{x \mid x = a_1 y_1 + a_2 y_2, -1 \leq y_1 \leq 1, -1 \leq y_2 \leq 1\}$

(b) $S = \{x \in \mathbb{R}^n \mid x \succeq 0, \mathbf{1}^T x = 1, \sum_{i=1}^n a_i x_i = b_1, \sum_{i=1}^n c_i x_i = b_2\}$, where $a_1, \dots, a_n, c_1, \dots, c_n \in \mathbb{R}$ and $b_1, b_2 \in \mathbb{R}$.

Yes. $S = \{x \in \mathbb{R}^n \mid -I x \preceq 0, \mathbf{1}^T x = 1, a^T x = b_1, c^T x = b_2\}$

Inequality: $-I x \preceq 0$

Equality: $Fx = g, F = \begin{bmatrix} \mathbf{1}^T \\ a^T \\ c^T \end{bmatrix} g = \begin{bmatrix} 1 \\ b_1 \\ b_2 \end{bmatrix}$

(c) $S = \{x \in \mathbb{R}^n \mid x \succeq 0, x^T y \leq 1 \text{ for all } y \text{ with } \|y\|_2 = 1\}$.

if $x \neq 0$, $\bar{y} = \frac{x}{\|x\|_2} \Rightarrow x^T \bar{y} \leq 1 \Rightarrow x^T \cdot \frac{x}{\|x\|_2} \leq 1 \Rightarrow \frac{\|x\|_2^2}{\|x\|_2} \leq 1$

$$\Rightarrow \|x\|_2 \leq 1$$

$$\Rightarrow x^T y \leq \|x\|_2 \cdot \|y\|_2 \leq 1$$

$$\Rightarrow S = \{x \in \mathbb{R}^n \mid x \succeq 0, \|x\|_2 \leq 1\} \Rightarrow \text{Ball not polyhedra}$$

(d) $S = \{x \in \mathbb{R}^n \mid x \succeq 0, x^T y \leq 1 \text{ for all } y \text{ with } \sum_{i=1}^n |y_i| = 1\}$.

$$x^T y \leq 1 \Rightarrow x^T \begin{pmatrix} 1 \\ \vdots \\ 0 \end{pmatrix} \leq 1 \Rightarrow x_1 \leq 1$$

Similarly. we can get $x_i \leq 1$

$$\text{Since } x \succeq 0 \Rightarrow 0 \leq x_i \leq 1$$

$\Rightarrow \text{box} \Rightarrow \text{polyhedra.}$

$$\Rightarrow \begin{bmatrix} -I \\ I \end{bmatrix} x \leq \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Question 7

Solution set of a quadratic inequality. Let $C \subseteq \mathbb{R}^n$ be the solution set of a quadratic inequality,

$$C = \{x \in \mathbb{R}^n \mid x^T A x + b^T x + c \leq 0\},$$

with $A \in \mathbb{S}^n$, $b \in \mathbb{R}^n$, and $c \in \mathbb{R}$.

(a) Show that C is convex if $A \succeq 0$.

$$\text{let } f(x) = x^T A x + b^T x + c$$

$$f'(x) = A x + A^T x + b^T = 2A x + b^T$$

$$f''(x) = 2A \succeq 0 \Rightarrow \text{convex}$$

(b) Show that the intersection of C and the hyperplane defined by $g^T x + h = 0$ (where $g \neq 0$) is convex if $A + \lambda g g^T \succeq 0$ for some $\lambda \in \mathbb{R}$.

Are the converses of these statements true?

$$\text{let } q(x) = x^T A x + b^T x + c \leq 0$$

$$\text{and } H = \{x \mid g^T x + h = 0\}, g \neq 0$$

$$\begin{aligned} \Rightarrow q(x) + \lambda H^2 \leq 0 &\Rightarrow x^T A x + b^T x + c + x^T (g g^T) x + 2h g^T x + h^2 \leq 0 \\ &\Rightarrow x^T (A + \lambda g g^T) x + (b + 2\lambda h g)^T x + (c + \lambda h^2) = w(x) \end{aligned}$$

if $A + \lambda g g^T \succeq 0$, then w is convex

On H , $C \cap H = \{x \in H \mid q(x) \leq 0\} = \{x \in H \mid w(x) \leq 0\}$ is convex

$\Rightarrow$ Intersection of convex sets is convex.

Converses:

① if $C = \{x \mid x^T A x + b^T x + c \leq 0\}$ is convex, must $A \succeq 0$?

When $n=1$, $A=-1$, $b=0$, $c=0$,

Then $C = \{x \mid -x^2 \leq 0\} = \mathbb{R}$ is convex, while $A=-1 \not\succeq 0$

$\Rightarrow$ Not True.

⑥ if $C \cap H$ is convex, must there exist λ such that $A + \lambda g g^T \succeq 0$.

$$\text{Let } A = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, b=0, c=0, n=2$$

$$\Rightarrow C = \{(x_1, x_2) \mid -x_1^2 + x_2^2 \leq 0\} = \{(x_1, x_2) \mid |x_2| \leq |x_1|\}$$

$\Rightarrow$ non-convex.

$$\text{let } H_2 \text{ be } x_2=0, \quad g = (0, 1), \quad h=0$$

$$\Rightarrow -x_1^2 + 0 \leq 0 \Rightarrow C \cap H = \{(x_1, 0) \mid x_1 \in \mathbb{R}\} \text{ is a line,}$$

hence convex.

$$\text{But } A + \lambda g g^T = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} + \lambda \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & 1+\lambda \end{pmatrix}$$

$\Rightarrow$ no such λ could fix the -1 .

$\Rightarrow$ NOT True